Comparative Assessment: MILU vs. MMLU

Deployment context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Deployment Context

Use case: Task: Environment science and geograpy-specific agricultural knowledge understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: Environmental science

Setting: Environment science research utility assessment using llm and local-context

Target population: An environment scientist from Mymensingh, Bangladesh wants to get knowledge about different south asian region specific agricultural practice. So having different region and language specific agricultural and demographic context understanding (e.g. telegu spoken area, hydrabad India and agricultural practice/problems vs farmers practice/problems from Mymensingh, Bangladesh~bengali spoken in Mymensingh accented dialect.)

Overall Risk Ratings

Benchmark	Risk	Avg. Score
MILU	HIGH	2.2 / 5
MMLU	HIGH	1.2 / 5

Dimension Scores Comparison

Dimension	MILU	Rating	MMLU	Rating	Δ
Input Ontology (IO)	2/5	Significant gaps	1/5	Serious concern	+1
Input Content (IC)	1/5	Serious concern	1/5	Serious concern	0
Input Form (IF)	3/5	Moderate gaps	1/5	Serious concern	+2
Output Ontology (OO)	2/5	Significant gaps	1/5	Serious concern	+1
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	3/5	Moderate gaps	2/5	Significant gaps	+1
Average	**2.2**		**1.2**		**+1**

Δ = MILU score − MMLU score. Positive = regional benchmark scores higher.

Overall Summaries

MILU (risk: HIGH)

MILU is fundamentally misaligned with the Mymensingh environmental-scientist deployment context. The benchmark is by design India-centric across all six validity dimensions: its 41-subject taxonomy lacks Bangladesh-specific agro-ecological subdomains; its content originates exclusively from Indian competitive exams with no Bangladeshi farming, dialect, or trans-border hydrology representation; its Bengali corpus reflects West Bengal exam register and is heavily machine-translated (100% of the sampled validation Bengali split); its annotation pool excluded Bangladeshi subject-matter experts; and its single-correct-answer MCQ format cannot accommodate the cross-border knowledge pluralism that trans-border water-policy questions demand. Dataset analysis empirically confirms these structural gaps: Bangladesh appears in 215 sampled examples only as a wrong-answer distractor, Agriculture-labeled questions concern Indian schemes and corporate trivia rather than agronomy, and Telangana/AP agro-ecological content is absent. MILU's strengths — broad Indic-script coverage, consistent MCQ structure, broad domain diagnostic breadth — make it useful as a generic Indic-LLM screening tool, but it provides essentially no discriminative signal for Bangladeshi agricultural and environmental science knowledge.

MMLU (risk: HIGH)

MMLU is fundamentally misaligned with the South Asian Agricultural and Environmental Science deployment across five of six dimensions. Five HIGH-priority dimensions (Input Ontology, Input Content, Input Form, Output Ontology, Output Content) score 1 — the lowest possible — with multiple structural validity violations: complete absence of agro-ecological subjects, US/Anglophone-saturated content, English-only input, MCQ schema unable to encode regional pluralism, and US-anchored ground-truth labels with no South Asian annotator validation. Output Form scores 2, partially supporting cross-model comparison via accuracy metrics. The single biggest concern is that the benchmark cannot, by construction, measure the constructs the user needs to evaluate — there is no item in MMLU's 14,079-question test set addressing haor wetland ecology, boro/aman/aus rice cultivation, char-land farming, Deccan dry-land cropping, or coastal aquaculture. Cross-border Bangladesh–India water-policy questions, which are actively contested in 2025, are entirely absent and could not be represented in the four-option MCQ format even if added. The benchmark is suitable only as a weak generic STEM-reasoning baseline for cross-model comparison.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

MILU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's 41-subject taxonomy includes an Environmental Sciences domain and an Agriculture subject, but both are surface-level and India-scoped by design [Q19, Q38]. The taxonomy lacks any Bangladesh-specific agro-ecological subdomains — no haor/beel wetland category, no char-land farming, no boro/aman rice cycle subject, and no trans-border water management category. Dataset analysis confirms this empirically: across 215 sampled examples, Environmental Sciences is populated almost entirely with geography trivia (DATASET-D6, D9, D16, D18) rather than agro-ecological science, and the few Agriculture-labeled questions concern Indian schemes/institutions (DATASET-D10, D13, D14). The authors themselves note poor model performance in 'culturally relevant areas' [Q6, Q68] which the deployment population would consider their core expertise. For a HIGH-priority deployment requiring agro-ecological depth across Bangladeshi delta and Indian cross-reference zones, the taxonomy is materially underrepresentative.	MMLU's 57-subject taxonomy is organized around US/Western academic disciplines [Q1, Q9, Q18] and contains no subject addressing agro-ecology, delta hydrology, haor wetland systems, boro/aman/aus rice cultivation, char-land farming, Deccan dry-land cropping, or coastal aquaculture — the precise subdomains required for this deployment. The dataset analysis confirmed this empirically across 175+ examples sampled from 37 configs: the closest environmental content observed is generic water chemistry and pastoralism definitions, with no South Asian agricultural systems represented [DATASET-D19, D22, D32]. Many subjects are actively irrelevant (US foreign policy, US history, US government), introducing construct-irrelevant variance for an agricultural science deployment. Given IO is a HIGH-priority dimension and the gap is structural and complete, this is a major validity violation.

MILU evidence	MMLU evidence
[Q19] 'MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge.' (p.3)	[Q1] 'The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.' (p.1)
[Q38] 'we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.' (p.4)	[Q9] 'The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more.' (p.1)

[Q6] 'Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM.' (p.1)	[Q18] 'The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn.' (p.3)
DATASET-D10: Only Bengali Agriculture-labeled question concerns India's National Food Security Mission, not Bangladeshi farming	DATASET-D19: high_school_geography pastoralism definition is closest agriculture-adjacent content; no South Asian system represented
DATASET-D13: Marathi Agriculture question asks location of National Mustard Research Centre (Sever, Rajasthan) — institutional trivia not agronomy	DATASET-D22: college_biology covers cellular/molecular content, not agro-ecology
DATASET-D14: Odia Agriculture question asks which Indian company makes farm equipment — corporate trivia	DATASET-D32: nearest 'environmental science' is generic water-pollutant chemistry, not delta/saline-intrusion or haor ecology
DATASET-D18: Environmental Sciences domain populated with Mediterranean climate trivia, not agro-ecology	
DATASET-D9: Environmental Sciences/Geography subject contains Telangana welfare-pension question — taxonomy applied loosely	

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

MILU — 1/5 (Serious concern) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
Every datapoint in MILU originates from Indian competitive exams [Q9, Q11, Q20, Q24, Q93]. The deployment YAML and web search confirm structural absence of Bangladeshi content (GAP-1, GAP-6 [WEB-17, WEB-18]). Dataset analysis confirms this empirically: across 215 sampled examples, Bangladesh appears only as a wrong-answer distractor in two questions about Indian Pharmacopoeia (DATASET-D21, D25), and no Bangladeshi institution, ecological system, or trans-border policy appears anywhere in the sample (CRITICAL Concern 1). Bengali content is West Bengal exam-derived, with the entire validation Bengali sample being machine-translated from English (DATASET-D3, all 21 examples). Telugu content shows no Telangana/AP agro-ecological specificity (MAJOR Concern 6). For a HIGH-priority dimension targeting Bangladeshi farming context, Mymensingh dialect terminology, haor wetlands, and trans-border hydrology, the content is essentially zero — a fundamental misalignment.	All MMLU content is sourced from US standardized tests (GRE, USMLE), undergraduate course materials, and Oxford University Press books [Q19], with Professional Law supplemented by Harvard Law Library case law [Q77] and history items drawing on English primary sources [Q140, Q141]. The authorship team is entirely US-based [Q10] and funding is from US sources [Q87, Q88]. Empirical sampling confirms US-centric content saturation: US tort law [DATASET-D7], US contract law [DATASET-D8], US Social Security [DATASET-D9], US birth control statistics [DATASET-D10], US Populist movement [DATASET-D2], MCAT references [DATASET-D39], and US measurement units [DATASET-D29]. South Asian content is limited to three tangential references (India internet 2017, India vs. Congo university graduates, Jainism/samsara) and one colonially-framed Indian Uprising item [DATASET-D13, D40, D36, D18]. No content from BRRI, BARI, BAU, ICAR, PJTSAU, or ANGRAU was identified, and gap searches confirmed no Bengali- or Telugu-language agricultural NLP corpus exists in the public landscape [WEB-22, WEB-6]. The Mymensingh-dialect agricultural vocabulary, Telugu dry-land terminology, and cross-border water-policy content required for this deployment are absent by construction.

MILU evidence	MMLU evidence

[Q9] 'We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.' (p.2)	[Q19] 'The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination.' (p.3)
[Q11] 'We create this benchmark by collecting questions from over 1500 competitive exams from India.' (p.2)	[Q77] 'We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus' (p.8)
[Q24] 'Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state.' (p.3)	[Q10] 'Dan Hendrycks, UC Berkeley; Collin Burns, Columbia University; Steven Basart, UChicago; Andy Zou, UC Berkeley; Mantas Mazeika, UIUC; Dawn Song, UC Berkeley; Jacob Steinhardt, UC Berkeley' (p.1)
[Q93] 'We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations.' (p.17)	[Q140] 'This question refers to the following information.' (p.19)
[WEB-17] AI4Bharat/MILU GitHub confirms exclusively India-centric exam sourcing	[Q141] 'English Parliament, Act of Supremacy, 1534' (p.19)
[WEB-18] NAACL paper confirms no Bangladeshi exam corpus	[WEB-22] AI4Bharat IndicNLP Catalog — no Bangladeshi-Bengali or Mymensingh-dialect agricultural corpus listed
DATASET-D21: Bangladesh appears only as wrong-answer distractor (Indian Pharmacopoeia question)	[WEB-6] ICICC 2024 Telugu agricultural perplexity study explicitly notes lack of Telugu agricultural NLP resources
DATASET-D25: Same Pharmacopoeia question in Telugu, Bangladesh again only as distractor	DATASET-D7: professional_law US tort scenario — confirms US legal context saturation
	DATASET-D8: professional_law US contract law — US cultural framing
	DATASET-D9: human_aging US Social Security — US institution as universal answer
	DATASET-D10: human_sexuality explicitly US-framed factual question
	DATASET-D29: elementary_mathematics uses miles per hour (US customary units)
	DATASET-D39: college_medicine references MCAT — US academic system
	DATASET-D13: global_facts India internet 2017 — only tangential South Asian factual reference
	DATASET-D40: global_facts India vs Congo university graduates — tangential
	DATASET-D36: world_religions Jainism — one of few South Asian cultural anchors
	DATASET-D18: high_school_world_history Cawnpore framed through British colonial journalism

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

MILU — 3/5 (Moderate gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's text-based MCQ format is technically compatible with the deployment's text-only modality, and Bengali and Telugu scripts are reliably rendered (DATASET strengths 1, 4). However, two concerns reduce alignment: (a) the exam-register MCQ phrasing differs from naturalistic professional-query style (elicitation A4), and (b) approximately 25% of questions are machine-translated [Q43], with the sampled Bengali validation split observed at 100% translated (DATASET-D3, CRITICAL Concern 2), introducing register artifacts and occasional pipeline quality issues including incomplete question stems (DATASET-D34, D35, D36 — MINOR Concern 8). For a MODERATE-priority dimension where script support is the floor requirement, MILU clears that bar but does not match the naturalistic register a Mymensingh domain expert would use.	MMLU is exclusively English-language, text-only, multiple-choice [Q8, Q21, Q45, Q46]. The HuggingFace metadata confirms monolingual English; every one of the 175+ examples sampled is English-only [DATASET-D2, D29, etc.]. The deployment elicitation explicitly requires Telugu, Bangladeshi Bengali (Mymensingh dialect), and Indian Bengali as query languages [elicitation A4, Q4]. Bengali script (বাংলা) and Telugu script (తెలుగు) are entirely unsupported. The paper's own authors acknowledge the limitation that 'many important concepts are conveyed mainly through other modalities' [Q64]. No transliteration, code-switching, or multilingual variant exists. Web research confirms no Bengali or Telugu agricultural-domain LLM evaluation infrastructure exists [WEB-5, WEB-17, WEB-18, WEB-25] and that LLMs perform meaningfully worse in Bengali than English [WEB-5]. This is an absolute, format-level mismatch that no in-format remediation can address. IF is a HIGH-priority dimension for this deployment, making this a fundamental validity violation.

MILU evidence	MMLU evidence
[Q43] 'Of the total 79K questions, only 25% of questions are translated from English' (p.4)	[Q8] 'We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings.' (p.1)
DATASET-D3: 100% of 21 sampled Bengali validation items are is_translated=True	[Q45] 'We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]."' (p.6)
DATASET-D34: Bengali antonym question with target word missing — translation pipeline quality issue	[Q64] 'many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format.' (p.7)
DATASET-D35: Bengali argument-evaluation question missing argument content — pipeline quality issue	[Q21] 'We collected 15908 questions in total' (p.3)
DATASET-D36: English question with truncated stem	[WEB-5] arXiv 2507.23248 — LLMs perform worse in Bengali than English; smaller models especially degraded
	[WEB-17] AI4Bharat IndicBERT covers Bengali and Telugu but not agricultural domain
	[WEB-18] IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark
	[WEB-25] IndicXTREME (ACL 2023) covers Bengali/Telugu for general NLU but no agricultural evaluation
	DATASET-D2: typical English-only question format with no provision for non-Latin scripts
	DATASET-D29: even basic numeric items use English with US customary units

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

MILU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's output ontology is single-correct-answer MCQ over 41 India-derived subject labels and 8 domains [Q35, Q36, Q37, Q38]. The structural single-correct-label format cannot accommodate cross-border knowledge pluralism on contested topics like trans-border water policy — the deployment YAML identifies this as a HIGH-priority concern (transboundary_policy_context, GAP-3 [WEB-11, WEB-14]). Even if such questions existed, the format would force a single nationally-partial answer key (MAJOR Concern 7: DATASET-D5, D28). The 41-subject labels themselves were derived by clustering ~20K Indian-exam tags [Q35, Q36] and validated against Indian standards. Domain-wise model performance is poor in culturally relevant areas [Q16, Q68], and the authors note a need for 'more inclusive data distribution' [Q70]. For a MODERATE-priority dimension, the structural rigidity plus India-derived label space creates meaningful misalignment, though the basic MCQ correctness structure is functional for non-contested factual content.	MMLU applies a uniform four-class MCQ schema (A/B/C/D) across all 57 tasks [Q23, Q35], with no mechanism to encode legitimate regional divergence in correct answers. The deployment explicitly anticipates such divergence — particularly for Bangladesh–India water-sharing questions [elicitation A3] where active 2025 geopolitical context (Ganges Treaty renewal, unresolved Teesta dispute, Kushiyara agreement) [WEB-1, WEB-2, WEB-4, WEB-21] would yield nationally divergent correct answers. The moral_scenarios config (500 items) explicitly anchors correctness to 'ordinary moral standards in the US as of 2020' as a constitutive framing across every item [DATASET-D1, D12], structurally precluding pluralism. The four-option forced-choice schema also cannot capture practice differences between delta flood-cycle planning and Deccan dry-land cropping. Models perform especially poorly on socially-important subjects (morality, law) [Q6, Q86] and on procedural/calculation tasks [Q14, Q72] — precisely the categories most relevant to agricultural decision-making. OO is HIGH-priority for this deployment, and the schema rigidity is a structural validity violation.

MILU evidence	MMLU evidence
[Q35] 'Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap.' (p.4)	[Q23] 'we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions.' (p.3)
[Q36] 'we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters.' (p.4)	[Q6] 'Worse, they still have near-random accuracy on some socially important subjects such as morality and law.' (p.1)
[Q38] 'we determine 41 distinct subject names, which fall into eight main domains' (p.4)	[Q14] 'The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality.' (p.2)
[Q16] 'Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM.' (p.2)	[Q86] 'Worryingly, models also perform especially poorly on socially relevant subjects including morality and law.' (p.8)
[Q70] 'Bridging this gap requires a more inclusive data distribution that ensures equitable representation of all cultures and languages.' (p.7)	[Q72] 'Models also have difficulty performing calculations, so much so that they exhibit poor performance on Elementary Mathematics' (p.8)
[WEB-11] Ganges treaty expires Dec 2026; renewal contested — bilateral ground-truth volatility	[WEB-1] Ganges Water Treaty expires Dec 2026 — bilateral water policy actively divergent
[WEB-14] No Teesta formal treaty exists — categorically contested ground truth	[WEB-2] Teesta dispute unresolved with no formal treaty — nationally-divergent correct answers likely
DATASET-D5: Indian Election Commissioner removal — Indian constitutional answer encoded as single correct response	[WEB-4] Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ

DATASET-D28: 44th Constitutional Amendment as definitive answer — Indian-specific legal fact	[WEB-21] 2022 Kushiyara Agreement — recent bilateral agreement requires up-to-date grounding
	DATASET-D1: moral_scenarios explicitly frames correctness as 'US as of 2020' — constitutive framing
	DATASET-D12: moral_scenarios US 2020 framing repeats across all five sampled items, confirming structural rather than incidental

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

MILU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
Ground-truth labels originate exclusively from Indian exam portals where 'subject experts ensure the accuracy of the answers' [Q22] and AI4Bharat volunteers conducted manual audits [Q87]. No Bangladeshi annotators, agronomists, or environmental scientists are documented in the annotation pool (GAP-6 resolution: 'CONFIRMED ABSENT' [WEB-17, WEB-18]). Approximately 45% of questions arrived pre-tagged from source portals, with the remainder receiving GPT-4O-MINI-generated topic assignments [Q33, Q34]. There is no documentation of inter-annotator agreement metrics or annotator demographics beyond AI4Bharat affiliation. The expert elicitation (A3) noted Indian exam answers are 'generally considered authoritative' for Bangladeshi environmental scientists, with notable exceptions around trans-border water agreements where Indian answers encode nationally partial perspectives — and the YAML documents that bilateral volatility (Ganges 2026 expiry, Teesta unresolved) makes this gap more acute [WEB-11, WEB-14]. Dataset analysis confirms Bangladesh appears only as wrong-answer distractor (DATASET-D21, D25) — Bangladeshi perspectives are structurally excluded. For a HIGH-priority dimension, this is a substantial convergent-validity concern.	The paper is silent on annotator demographics, selection criteria, regional or disciplinary backgrounds, or inter-annotator agreement [Q19]. The only documented human reference is US Amazon Mechanical Turk workers (34.5% accuracy) [Q22], with expert-level estimated from 'real-world test-taker human accuracy at the 95th percentile' on US standardized exams. The HuggingFace dataset card tags annotations_creators as 'no-annotation' [DATASET Concern 8]. The authorship team is entirely US-based [Q10] and content is sourced from US/Anglophone institutions [Q19, Q77]. There is no evidence — and no plausible mechanism — for South Asian agronomic, Mymensingh-dialect, or Telugu-region expert validation of any answer key. The user explicitly confirmed [elicitation A3] that cross-border water/agricultural questions could yield divergent correct answers and that practice differences between regions are not reflected in answer keys. Empirical sampling confirmed multiple items where US-specific facts (Social Security age increases, US birth control prevalence, US 2020 moral norms) are encoded as universally correct [DATASET-D1, D9, D10, D11, D12]. OC is HIGH-priority and represents a fundamental convergent and external validity violation.

MILU evidence	MMLU evidence
[Q22] 'These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers.' (p.3)	[Q19] 'The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online.' (p.3)
[Q33] 'approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information.' (p.4)	[Q22] 'Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test.' (p.3)
[Q34] 'we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question.' (p.4)	[Q10] 'Dan Hendrycks, UC Berkeley; Collin Burns, Columbia University; Steven Basart, UChicago...' (p.1)
[Q87] 'We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits.' (p.9)	DATASET-D1: moral_scenarios US 2020 framing as constitutive answer criterion
[WEB-11] Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge	DATASET-D9: human_aging Social Security as universally correct answer

[WEB-14] No formal Teesta treaty; positions categorically differ	DATASET-D10: human_sexuality US birth control statistics as correct answer
[WEB-17] AI4Bharat/MILU GitHub: no Bangladeshi annotator participation	DATASET-D11: human_sexuality 1988-1990 US-specific survey data as correct answer
[WEB-18] NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only	DATASET-D12: US 2020 moral framing replicated across all sampled moral_scenarios items
DATASET-D21: Bangladesh appears only as wrong-answer distractor in Pharmacopoeia question	
DATASET-D25: Same Pharmacopoeia question in Telugu — Bangladesh as distractor across languages	

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

MILU — 3/5 (Moderate gaps) [high confidence]	MMLU — 2/5 (Significant gaps) [medium confidence]
MILU evaluates via accuracy on MCQ tasks using log-likelihood scoring for non-API models [Q48–Q51] and structured-JSON generative scoring for API models [Q53]. Both deployment and benchmark are text-based, satisfying the basic modality match. The MCQ output format does not require speech, vision, or other modalities that would create infrastructural barriers in Mymensingh (high text-only connectivity [WEB-20]). Two limitations reduce the score: (a) the authors acknowledge log-likelihood may yield different results than generation-based or CoT methods [Q85], which matters for the deployment's reasoning-heavy queries; and (b) MCQ output cannot capture the multi-step open-ended agricultural reasoning the deployment scenario requires (CRITICAL Concern 3, MAJOR Concern 7). For a LOWER-priority dimension where text modality is the main requirement, MILU is functionally adequate but suppresses richer output forms relevant to professional knowledge work.	MMLU evaluates only text-based MCQ classification accuracy [Q23, Q35] via predicted A/B/C/D tokens [Q45]. The metric (accuracy) and calibration measures (RMS calibration error, accuracy-confidence gap) [Q60, Q62] partially align with the deployment's stated need for cross-model comparison. However, the format cannot evaluate open-ended agronomic reasoning, does not penalize language mismatches (e.g., model responding in English to a Bengali prompt), and does not support regional-language outputs. The user did not flag output form as a primary concern (MODERATE priority), and MCQ accuracy is at least usable as a baseline. But the format precludes evaluating practice-difference reasoning, generative explanations of flood-cycle decisions, or Bengali/Telugu-language extension communication — all relevant for downstream farmer/agronomist applications. Combined with the 'esoteric knowledge' data bottleneck [Q81] amplified for South Asian agricultural content, output form is partially aligned but insufficient for deployment-validity evaluation.

MILU evidence	MMLU evidence
[Q48] 'For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.' (p.5)	[Q35] 'To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.' (p.5)
[Q53] 'We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.' (p.5)	[Q45] 'The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction.' (p.6)
[Q85] 'our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.' (p.9)	[Q60] 'We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.' (p.7)
[WEB-20] Bangladesh internet/mobile coverage adequate for text modality	[Q81] 'Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations.' (p.8)